Problem Chosen	2021 MCM/ICM Summary Sheet	Team Control Number
D		**2118577**

EMCA: A Framework for Studying Music Evolution and Cultural Changes

Summary

Music has been a part of people's lives since ancient times. With shifting characteristics, music has been on the course of evolution all through the history. To understand the factors that contribute to this phenomenon, we propose a novel **EMCA framework** to analyze the music evolution from the perspective of music characteristics, artists and genres.

Firstly, we apply **PageRank (PR)** and **Centrality Measurements** to quantify the music influence with given data. We also calculate the influence of each genre based on this benchmark. Additionally, a **subnetwork** is selected to exemplify the meaning of the value.

Secondly, we use **Cosine Correlation Coefficient** to measure music similarity. Using this method, we calculate the similarity distribution of artists within genres and across genres. The results show that the similarity of artists within a genre is higher than that of artists across genres, i.e. artists within a genre are more similar compared with the other group. Then, we use **Cluster Analysis** on music and discover that the result of Cluster Analysis is in good accordance with the real genre classification. Therefore, we contend that genres can be classified by music features.

Thirdly, we consider that there is an influencer-follower relationship between artists whose music similarity exceeds 0.9. Based on this, about 68% of the influence relationship is authentic. We further establish the D **index** to describe the gap of music features, and speculate that the feature with the smallest gap between influencers and followers is the most influential feature. The results show that speechiness and loudness are the most contagious music features. Next, we build a **Decision Tree Model** for classification. In the process of supervised learning, we have grasped the importance of various music features in the classification of genres: the importance of acousticsness, danceability, and instrumentalness ranks in the top 3, and therefore these three music features are the vital factors to distinguish genres. In addition, we use **Pearson Correlation Coefficient** to study the correlation between genres.

Finally, we build a **Time Series Model** to study the evolution of music and analyze the changes of music characteristics of different genres over time. Based on the Time Series Model, we examine the popularity of various genres over time, and analyze the changes in culture, society, and technology through visualization.

Moreover, our EMCA framework clearly manifests how to predict the evolution of music. It can be applied to make further simulation and prediction in any field, e.g. literature, history, etc., if data is provided.

Keywords: Directed Network of Music Influence, Decision Tree Model, Cluster Analysis, Pearson Correlation Coefficient, Cosine Correlation Coefficient, Time Series Model, Music Evolution

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1 Introduction

1.1 Restatement of the Problem

Music is often considered as essential and influential in terms of culture, society, and ideology, and therefore has played a significant role of all time. Although the heterogeneity of forms is acknowledged to all, music still possesses the quality of contagiousness. Artists can be greatly influenced by their predecessors in the process of creation, resulting in songs with similar characteristics. However, highly innovative artists, some major events or external changes in society might bring about musical evolution.

In order to better understand the course of evolution and revolution in music, the Integrative Collective Music (ICM) Society is currently planning to develop some models to measure musical influence and examine evolutionary and revolutionary trends of artists and genres. To make it clearer and well organized, we divide the given problem into the following tasks:

- Task 1: Develop a directed network model of music influence to determine the influence of artists and genres;
- Task 2: Create a musical similarity model to compare the similarity of artists within and between genres to verify the objective existence of influence between artists; analyze the contagiousness of different musical characteristics;
- Task 3: Build a decision tree model to identify some important musical characteristics in genre classification and analyze the similarity, correlation and mutual influence between genres;
- Task 4: Establish a time series model to explain the evolution of music genres over time, figure out certain turning points, influential artists, and genres in music development, and identify the effects of particular social, political, and technological changes.

1.2 EMCA Framework

Previous study has shed light on the process of music evolution and revolution: Matthias Mauch et al. [1] studied the evolution of American pop music from 1960 to 2010; Hemant Kumbhar et al. [2] proposed the genetic evolution theory to study music revolution. However, the above works only focused on the music evolution of a certain genre or the whole

music industry, and did not take into account the influence of the interaction between different music genres or artists on the music evolution.

Therefore, in this task, we devise a research framework named EMCA (Establishment, Measurement, Classification, and Analysis) to fulfil the aim of measuring the influence as well as exploring the evolution of music, which has a certain guiding significance for the study of music evolution and revolution. The details of the EMCA framework are shown in Figure 1.

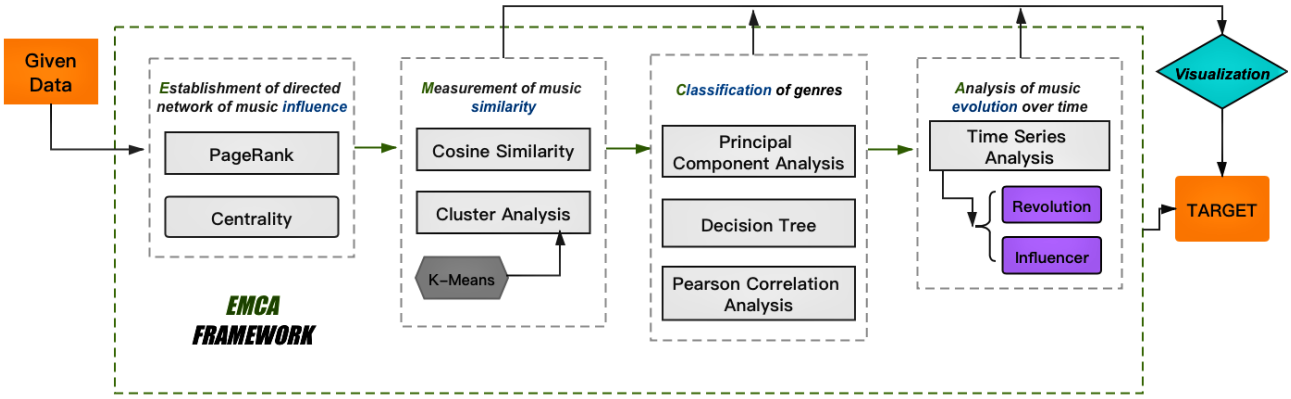

Figure 1: Framework of EMCA (Establishment, Measurement, Classification, and Analysis of Music Influence and Evolution)

The specific procedures of the EMCA framework are as follows:

- **Establishment:** We establish a "music influence" oriented network, using social network model to measure music influence by the value of *PageRank*, *Betweenness-Centrality*, *Closeness-Centrality*, and *Degree-Centrality*.
- **Measurement:** We utilize the *Cosine Correlation Coefficient* of music feature vector to measure the similarity between music. After that, we use the method of *Clustering Analysis* (K-Means) to test the rationality of this benchmark.
- **Classification:** By means of the *Decision Tree Model*, we have a clearer classification of music genres. Then, we calculate the *Pearson Correlation Coefficient* between different genres for correlation analysis. Through the method of *Principal Component Analysis*, we acquire some significant music characteristic factors that distinguish different genres from others.
- **Analysis:** We explore the evolution of music through *Time Series Analysis*, and seek for the critical turning points and major influencers in the process of music development.
- **Visualization:** We apply visual effects to make the results of EMCA more directly perceived.
- **Further Study Recommendation:** Since the two data sets provided are limited to certain genres and artists, we put forward further solutions for situations where richer data is available.

2 Assumptions

After adequate analysis of the problem, we make some general assumptions to simplify our model, as listed below:

Assumption 1. All the music produced by an artist within a certain genre belong to that genre.

Assumption 2. We consider the music with a cosine similarity of 90% or more of the music characteristics as highly similar, i.e. when two artists have a music similarity of 90% or more, they are considered to have an influencer-follower relationship.

Assumption 3. The influence on the follower is considered to be instantaneous, i.e. the time of the influence process is neglected.

Assumption 4. The genre that an artist belongs to does not change over time.

Assumption 5. An artist's influence begins in the year when his or her career starts.

Other assumptions based on different models will be justified in the following model-related sections.

3 Notations

For compactness, we define a series of symbols for our models in Table 1.

Table 1: **Notations**

Symbol	Definition
t	Time (year)
$I(t)$	The influence of certain artist or genre of year t
CS	Music Cosine Similarity
PC	The Pearson Correlation Coefficient across genres
MV_i	The music feature vector of music i after normalization
GV_i	The music feature vector of genre i after normalization
D_i	The difference of music characteristics i
$P(t)$	Music popularity of year t
$PR(t)$	The PageRank of year t
$BC(t)$	The Betweenness-Centrality of year t
$CC(t)$	The Closeness-Centrality of year t
$DC(t)$	The Degree-Centrality of year t

4 Directed Network of Music Influence

We use the *influence_data* data set to develop a Directed Network of Music Influence so as to quantify the influence of artists and genres. In this network, influencers and followers are closely connected, forming a complex topological structure similar to that of a social network. On the basis of the Social Network Influence Theory [4], we comprehensively measure the musical influence of each artist by calculating PageRank (PR), Betweenness-Centrality (BC), Closeness-Centrality (CC), and Degree-Centrality (DC) of each node of the network. PR ranks the significance of an artist by the link relations of a substantial amount of them. DC examines the accurate number of followers of an artist. BC measures the centrality of an artist, i.e. the frequency of passing through this node to reach another one in this network. CC indicates whether the artist is in close proximity to other artists in this network—if one is close to all other nodes, it can be considered as central in spatial position.

The artists's influence is measured by the equation

$$I(t) = \frac{PR(t)}{2} + \frac{BC(t) + CC(t) + DC(t)}{6} \tag{1}$$

This measure combines PageRank and Centrality to reach an all-sided consideration, balancing the extreme result which occurs when measured by PageRank alone. For instance, if a distinguished artist follows a less-known one according to personal statement or expert opinion, this will produce a comparatively high PR result for the latter artist, which might be exaggerated in authentic situation. Figure 2 shows the result of PR while Table 2 examplifies the top 6 artists as the result of BC, CC, and DC. To further illustrate how we measure influence, we have taken a subnetwork from the above directed network model to make it more clearly, as manifested in Figure 2. This is an examplification of the massive network where thousands of artists interact with each other.

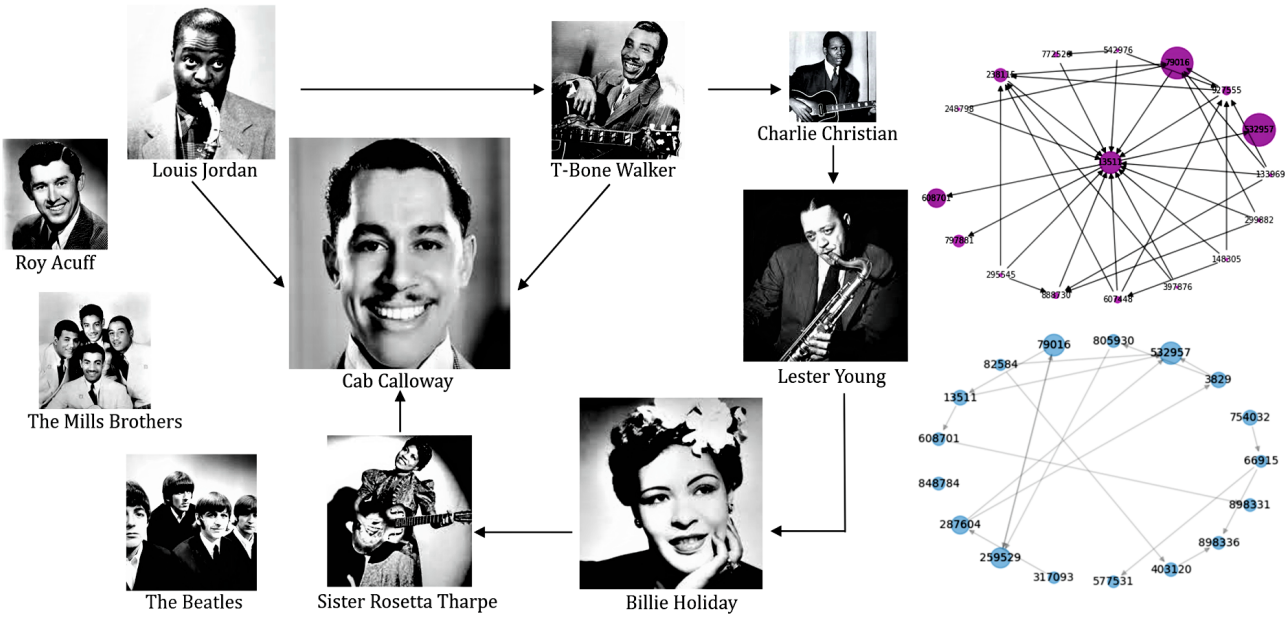

Figure 2: Influence of Artists–PageRank Result and Subnetwork Diagram

Table 2: BC, CC, and DC Result

	Betweenness-Centrality	**Closeness-Centrality**	**Degree-Centrality**
1	The Beatles (Pop/Rock)	Bob Dylan (Pop/Rock)	The Beatles (Pop/Rock)
2	Bob Dylan (Pop/Rock)	The Beatles (Pop/Rock)	Bob Dylan (Pop/Rock)
3	Rolling Stones (Pop/Rock)	Chuck Berry (Pop/Rock)	Rolling Stones (Pop/Rock)
4	Jimi Hendrix (Pop/Rock)	James Brown (R&B)	David Bowie (Pop/Rock)
5	Miles Davis (Jazz)	Little Richard (Pop/Rock)	Led Zeppelin (Pop/Rock)
6	David Bowie (Pop/Rock)	Elvis Presley (Pop/Rock)	Jimi Hendrix (Pop/Rock)

From these results, we come to the conclusion that the Beatles rank relatively high on all four measures of musical influence. This accords closely with the history of Western music in the twentieth century. People used to have separate music propensity, but in the mid-fifties, just a few years before the Beatles entered the public eye, the American music market experienced an integration with different music genres blending together, creating a new type of music–Rock n' Roll. The Beatles act as a lead character of this genre, and have been ravingly worshipped by the young adults. Overall, as we probe into the twentieth century's remarkable achievements, the Beatles are predictably prominent across a variety of perspectives. [3] Furthermore, most of the top 10 musicians on PageRank's list began their musical careers in 1930.

Individual artists play an important role in the network of musical influence; that is to say, they have great impact on the evolution of music. However, from a macro perspective, the evolutionary process of music still takes place mainly in terms of genres, which rise and fall in different eras. There are diverse mainstream music in different periods, which to a large extent influence the development of music at that time. For this reason, in addition to the influence network of musicians, we also build the influence network of music across genres and calculate the influence of each genre. The result is shown in Figure 3. Pop/Rock genre is undoubtedly the most influential one, followed by Jazz and R&B.

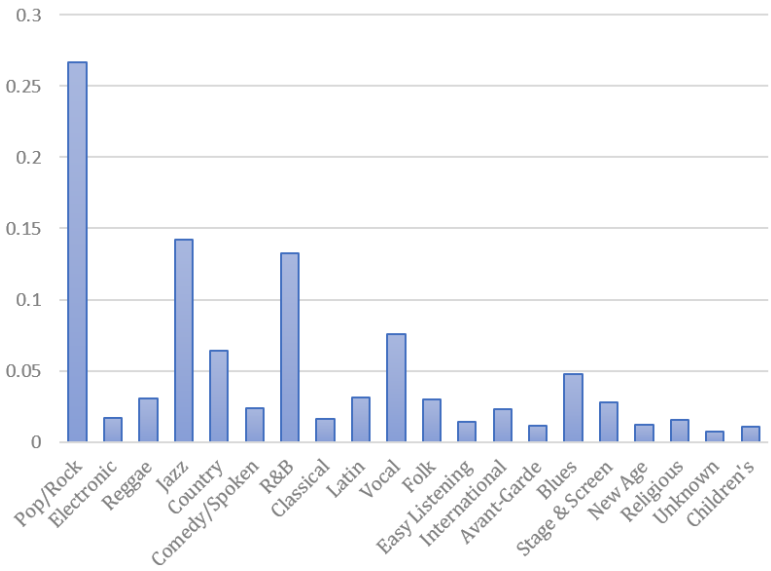

Figure 3: Influence of Genres

5 Measurement of Music Similarity

After establishing the Directed Network of Music Influence to quantify the influence of artists, it is necessary to probe into the similarity of songs and examine whether the influencers actually affect the music created by the followers.

5.1 Cosine Similarity

First of all, we normalize given music features to ensure that they can be processed together. We use

$$MV_i = \frac{V - Max}{Max - Min} \tag{2}$$

where V stands for the vector before normalization; Max represents the vector composed of the largest values in each dimension while Min represents the vector composed of the smallest values.

After normalizing all music features, we measure the similarity between music by finding the Cosine Correlation Coefficient of the music feature vector, i.e.,

$$\begin{aligned} PC &= \frac{cov(GV_x, GV_y)}{\sigma_x \sigma_y} \\ &= \frac{E(GV_x * GV_y) - E(GV_x)E(GV_y)}{\sqrt{E(GV_x^2) - E^2(GV_x)}\sqrt{E(GV_y^2) - E^2(GV_y)}} \end{aligned} \tag{3}$$

With this Cosine Similarity Model, we calculate the similarity of artists within the same genre versus the similarity of artists across different genres. Take vocal genre as an example, we measure the similarity of two random artists in this genre and compare it with the similarity of two random artists across any genre. The result, manifested in Figure 4, shows that the mean value of music similarity within the vocal genre is about 0.935, while the mean value of music similarity across different genres is lower than 0.9. Additionally, in terms of overall distribution, the value of the vocal group appears closer to 1.0 and more concentrated than the mixed group. Therefore, we draw the conclusion that the similarity of artists within the same music genre is higher than the similarity of artists between different genres.

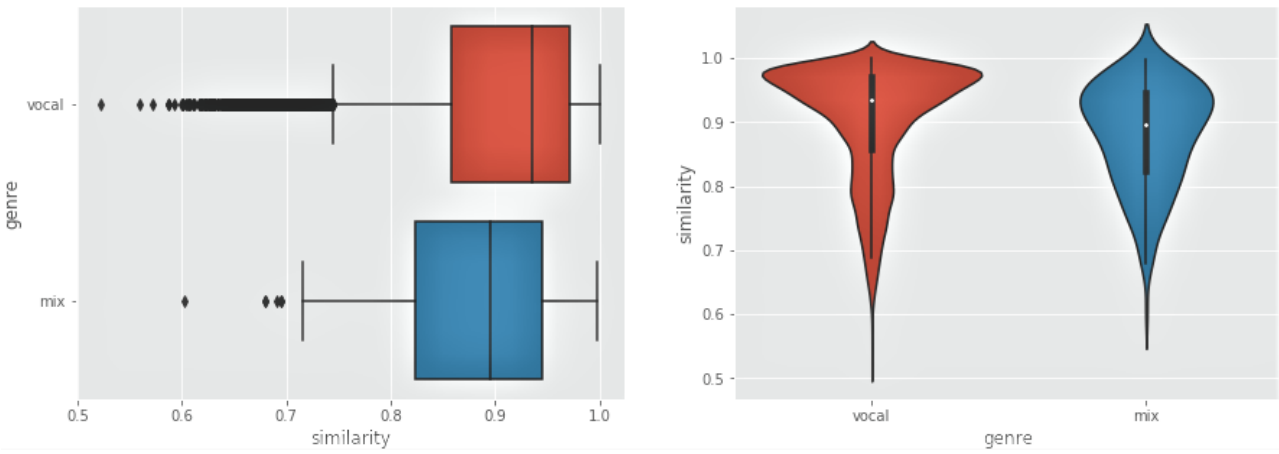

Figure 4: Comparison of Music Similarity Within and Across Genres

5.2 Cluster Analysis of Music Features

To investigate the relationship between music characteristics and genres, we first conduct an unsupervised learning process. We apply the K-Means method and extract the music features of each song as the feature vector. Then, Cluster Analysis is performed. We first obtain the genre of each song based on the genre of that artist; afterwards, in order to simplify calculation and classification, we select Jazz, Country, and Vocal songs as the input data for K-Means to perform the Cluster Analysis. We randomly choose two music features, acousticness and danceability, as the horizontal and vertical coordinates to depict the position of each point. According to our inference, the result of K-Means will not be so different from the actual result if the music features and genre classification have a certain relationship. The following Figure reffigure5 and Figure 6 are the actual distribution result and K-Means result of the songs respectively from these three genres on the two-dimensional plane. We find that the distribution of the point sets of different colors in these two figures are roughly the same.

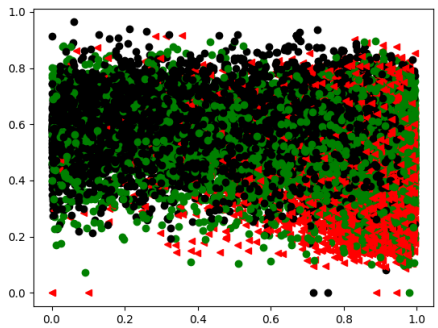

Figure 5: Actual Distribution Result

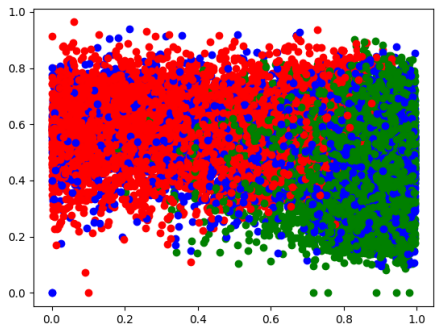

Figure 6: K-Means result

Actually, there is no strict quantitative standard about the classification of music genres for a long time. Heretofore, most of the music genres were categorized by composition techniques, instruments, or simply the feelings a song brings to people. Setting a way of classification through music characteristics is less frequent. In our study, the consistent result of Cluster Analysis of music characteristics and actual genre classification makes it possible to achieve quantitative genre categorization. Classifying music by its own acoustic characteristics makes it easier to organize music libraries on the Web, and this also reduces the controversy of genre classification.

5.3 Verification of Existence of Influence

Although we quantify the influence of artists and figure out the relationship of several noted artists, whether the influencer really affect the follower still require verification. We assume that if the interaction actually exists, the music composed by the follower should be more similar to that composed by the influencer. Based on similarity measurement, we obtain the distribution of music similarity between artists in *influence_data* data set.

Results shows that the overall music similarity among artists with certain relationship is higher than that among artists without relationship. The average is 0.93. In other words, the effect is objective; however, since music similarity within the same genre shows a high value without mutual influence, in order to investigate whether the influence is derived

from individuals or genres, we set a threshold of -0.9 for similarity (see Assumption 2). If the music similarity is above 0.9, it indicates that individuals will interact with and influence each other; if the music similarity is below 0.9, there might be no direct influence between individuals.

We single out influencers and followers with a music similarity of 0.9 or more on the basis of the original followers' relationship and consider that only these artists have the influencer-follower relationship. Among those singled out, we investigate the main factors that lead to the existence of a influencer-follower relationship. In other words, we probe into more contagious music characteristics that can attract more followers.

We contend that if followers are influenced because of certain contagious music characteristics, they will create music more similar to the works by the influencer in terms of those characteristics. According to this, we develop a model to measure the contagiousness of music characteristics, i.e.,

$$D_{feature} = \frac{|x_{feature} - y_{feature}|}{|Max_{feature} - Min_{feature}|} \tag{4}$$

$D_{feature}$ portrays the difference between music characteristics. The higher the degree of proximity shows, the greater the contagiousness of the feature is. We calculate $D_{feature}$ for each music charateristics, as listed in Figure 7.

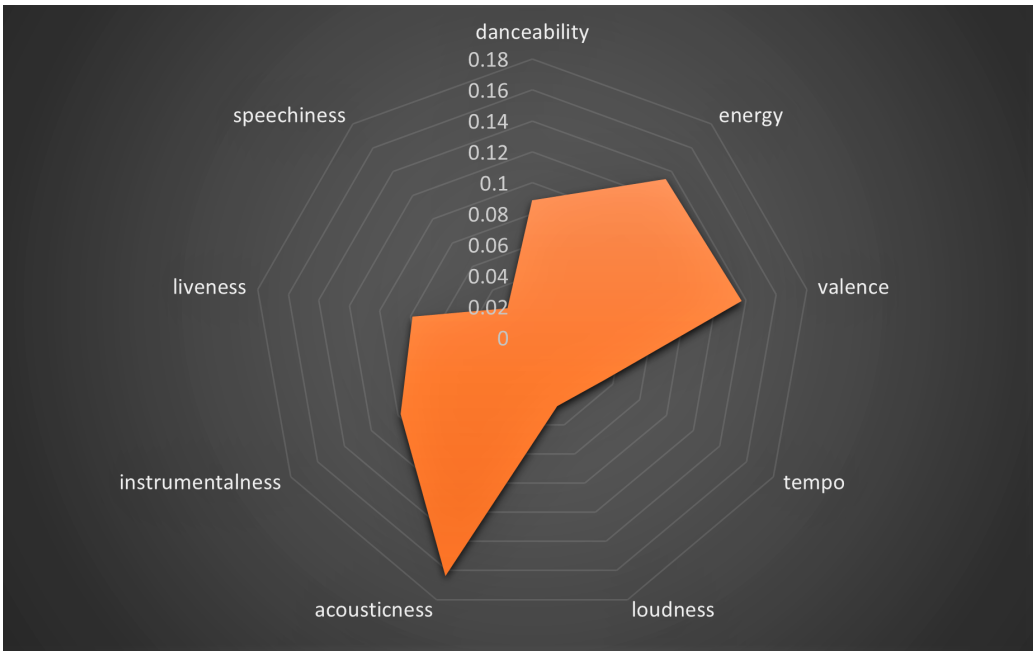

Figure 7: Average Percentage of Characteristic Changes

It is clear that $D_{feature}$ of speechiness and loudness features are smaller, which means that the music by influencer and follower are more similar on account of these two features. Therefore, speechiness and loudness features are more contagious and more likely to influence music composition.

6 Classification of Genres

6.1 Decision Tree Model

From the above study, we find that the musical characteristics seem to be somewhat related to the genre of a song. In order to receive a more valuable result out of the fact that the music characteristics within the same genre appear to be more similar, we decide to look deeper into the specific music characteristics that influence the classification of genres. If it is possible to distinguish the genre of a song by its music characteristics, this might become a more accurate and reliable way than manual classification.

A decision tree model is suitable for the problem of classification with multiple features. On the basis of a song as a unit, we regard musical features such as danceability, energy, acousticness and so on as the feature vector and the genre of the artist as the label of the song for supervised learning. Notably, features such as loudness need to be normalized in advance. We expect to visualize the model to further understand the relationship between music characteristics and genres.

Firstly, after building the decision tree model, we acquire the percentage of importance of each feature in the process of classifying genres. In Figure 8, we find that acousticness, instrumentalness, and danceability play a significant role in affecting the genre of the song, while the rest of the features possess a comparatively small ratio of less than 0.1. This result is reasonable, for features such as mode and key have their fixed types, and therefore are less decisive for classification. Similarly, features such as loudness do not have high ratio because they individually depend on the particular song. Moreover, this conclusion gives our group a hint: in the subsequent discussion of music characteristics, we can focus on acousticness, instrumentalness, and danceability which can be regarded as "decisive factors".

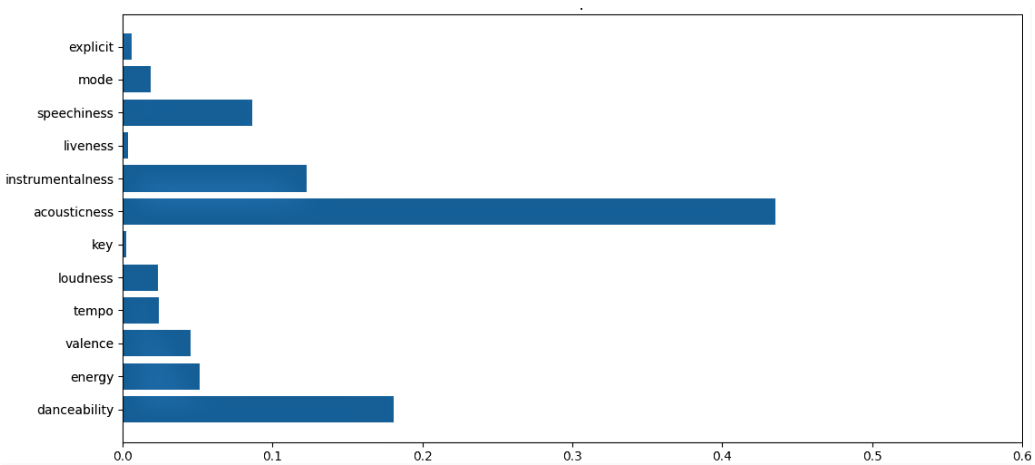

Figure 8: Importance of Different Music Features

In addition, according to visualization of the Decision Tree Classification Model, we can know exactly how the Decision Tree Model performs classification and further discover the difference in music characteristics between different genres.

When we set the maximum depth of the Decision Tree to 4, the result is as follows in Figure 9. Only 4 categories, namely Pop/Rock, Jazz, Classical, and Vocal can be identified. The result shows that when the acousticness is higher than 0.616 while the instrumentalness is higher than 0.085, the music is more likely to be Jazz. However, when the acousticness is higher than 0.616 while the instrumentalness is higher than 0.085, the music with lower

danceability is more likely to be Classical. To sum up, Jazz possesses higher danceability and instrumentalness, Classical possesses only high instrumentalness but low danceability, Vocal possesses very high acousticness, while Pop/Rock has a wider distribution of music characteristics.

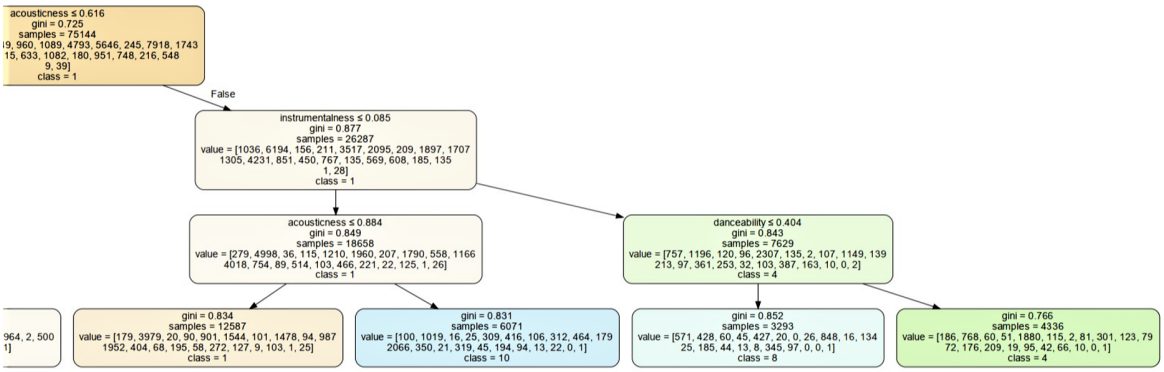

Figure 9: Decision Tree Example 1

By further increasing the maximum depth of the Decision Tree, as in Figure 10, the variety of music genres that can be identified will increase accordingly. However, it is disappointing that only when the Decision Tree is increased to about 11 as its maximum depth can all genres be identified, but the size of an 11-layer Decision Tree is already exceptionally large and complex.

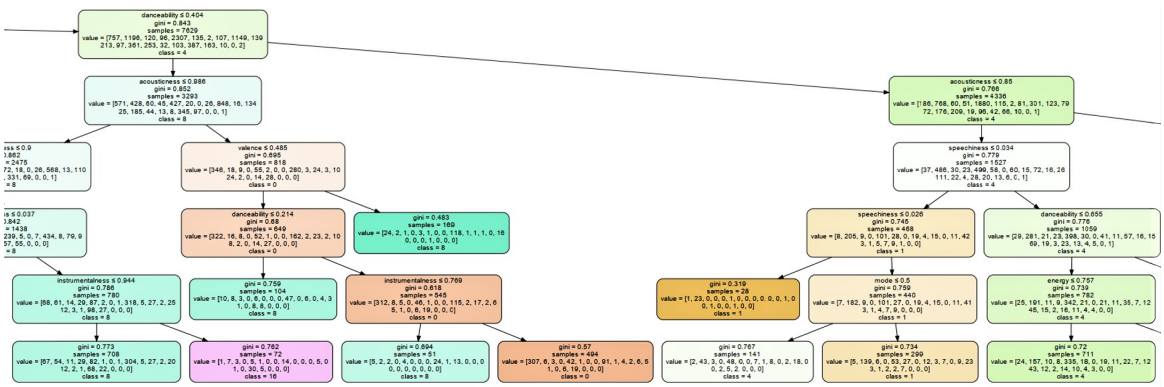

Figure 10: Decision Tree Example 2

From the Decision Tree Model, we come to know that not all features are important in classifying genres, and not all genres have their distinct feature in terms of music characteristics. From our data, the four genres, namely Pop/Rock, Jazz, Classical, and Vocal, can be better identified by music characteristics, i.e. these four genres are distinctive in music, while others are more variable and difficult to be analyzed by pure data.

6.2 Pearson Correlation Analysis

There are not only differences but also similarities between different music genres. In task 2, we found that the similarity between artists of the same genre is higher than the similarity between artists of different genres. Here, we consider genres as a whole and analyze the similarity as well as the degree of correlation across genres.

The first step is to obtain the feature vector of each genre. We take the average of every music feature of all songs from each genre as the value of that genre on the corresponding feature. After that we apply Pearson Correlation Coefficient:

$$\begin{aligned} PC &= \frac{cov(GV_x, GV_y)}{\sigma_x \sigma_y} \\ &= \frac{E(GV_x * GV_y) - E(GV_x)E(GV_y)}{\sqrt{E(GV_x^2) - E^2(GV_x)}\sqrt{E(GV_y^2) - E^2(GV_y)}} \end{aligned} \tag{5}$$

to calculate the similarity of music across genres. We draw a Heat Map in Figure 11 to visualize the result.

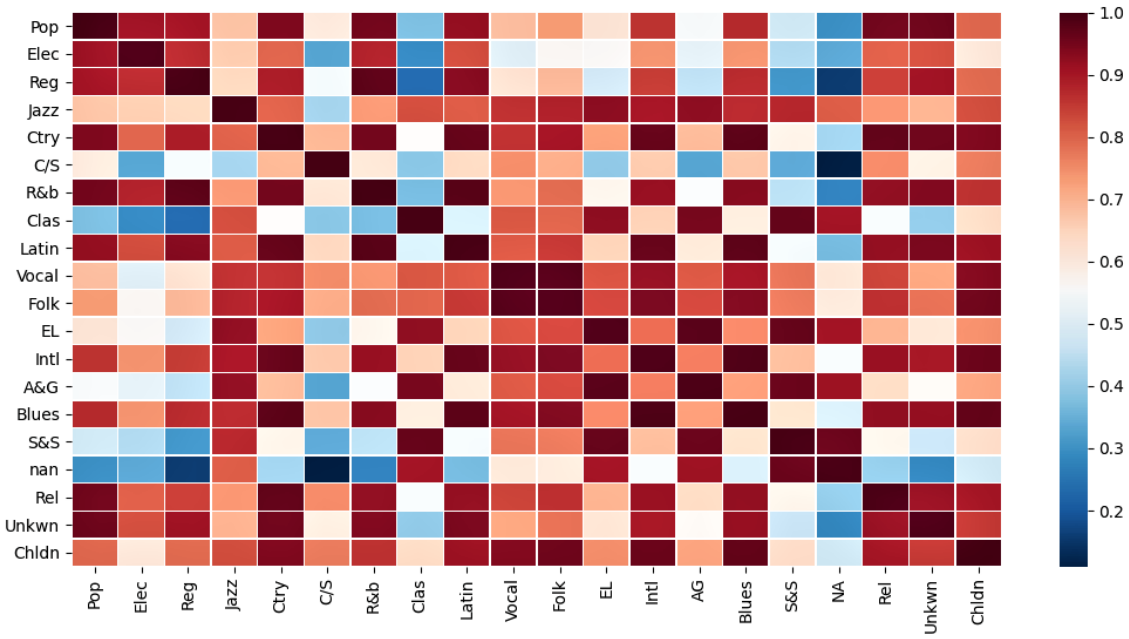

Figure 11: Heat Map of Genre Similarity

It can be seen that the map is irregular with alternating red-and-blue, which indicates that there are very similar or distinctive relationships across genres. Overall, there are more red squares than blue squares in the graph, so it is presumable that genres appear to be more alike than different.

We find that the four groups of genres in Table 3 below are highly correlated, with the Pearson Correlation Coefficient higher than 0.98. This indicates that in each group of two genres, there is little difference in terms of music characteristics. We then speculate that one genre might be derived from another, or two genres will continue to interact or even merge into one genre over time.

Table 3: 4 Groups of Similar Genres

Genre 1	Genre 2
International	Blues
Vocal	Folk
Easy Listening	Avant-Garde
R&B	Latin

7 Analysis of Music Evolution

7.1 Time Series Model

From the Decision Tree Model, we seek out some significant features for genre classification: danceability, acousticness, instrumentalness. Hence, when studying the evolution of music genre, we will mainly analyze the changes of these three music characteristics over time in the genre of R&B, Jazz, Pop/Rock,and Country. As for why these four genres are chosen, these four genres have a large amount of song data to analyze so as to ensure the reliability of analysis compared to other genres; on the other hand, the analysis of some certain genres is enough because we have mentioned above that there is a large degree of similarity between genres. In short, we contend that analyzing three music charateristics of four genres is sufficiently representative.

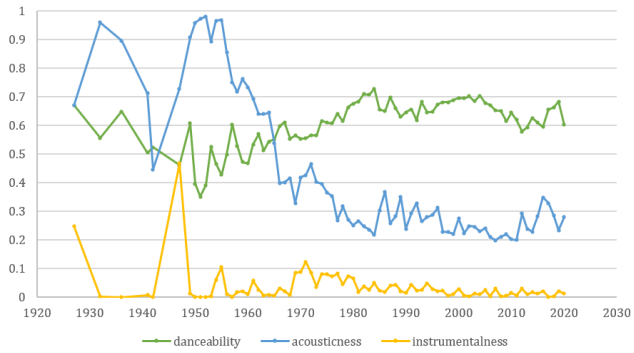

Figure 12: R&B Characteristics By Year

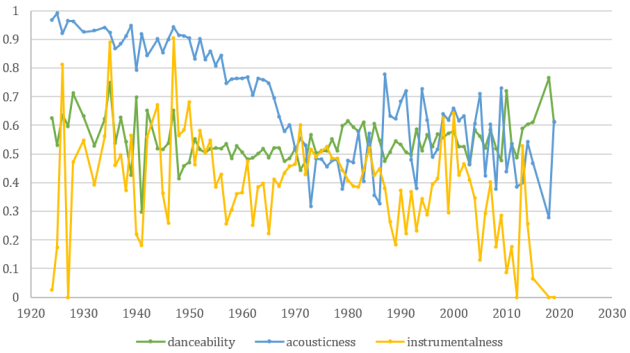

Figure 13: Jazz Characteristics By Year

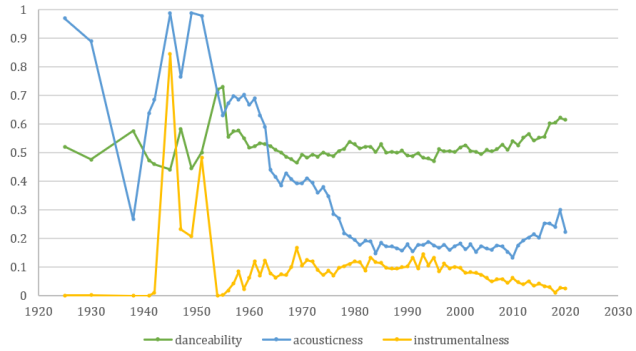

Figure 14: Pop/Rock Characteristics By Year

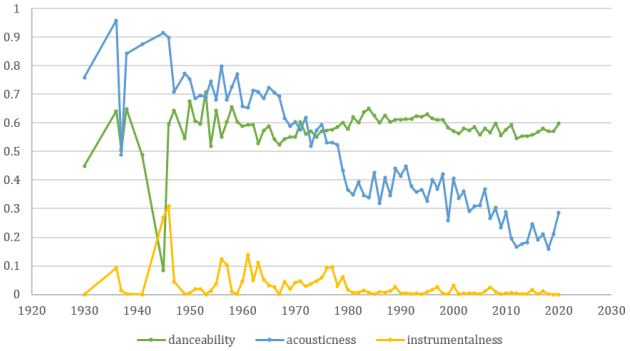

Figure 15: Country Characteristics By Year

The result is in Figure 12-15. In these four graphs, we can see a common trend: in the 1950s, there was a "sudden change" in the character of music: those characteristics either suffered a sharp decline or possessed a vigorous rise. It is assumed that during this time period

there was a "revolution" in music and that all genres were seeking change. After this period, however, the music characteristics of all genres, except for Jazz, leveled off, and therefore the changes were not particularly significant. This leads us to the general conclusion that after the 1950s, when something extraordinary happened that revolutionized the music field, there was no such seismic shift any more, and the music developed in a steady manner. However, Jazz is an exception as it has always held a significant level of change. This explains why music characteristics cannot be used exclusively to classify genres— sometimes they're always changeable.

7.2 Music Revolution

In order to find out music revolution point, we process the data of popularity. According to Figure 16 and Figure 17, there was a big increase in the popularity of music in the 1950s, especially between 1955 and 1956, which can be regarded as a revolution.

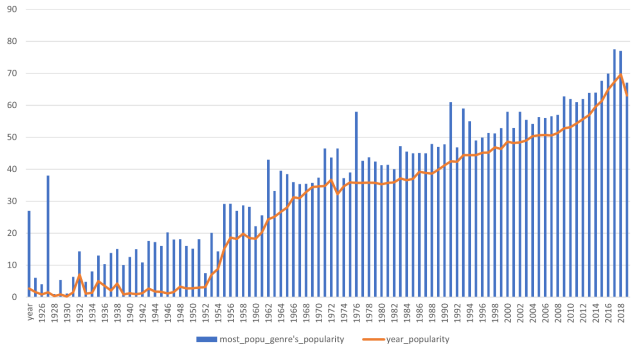

Figure 16: Popularity

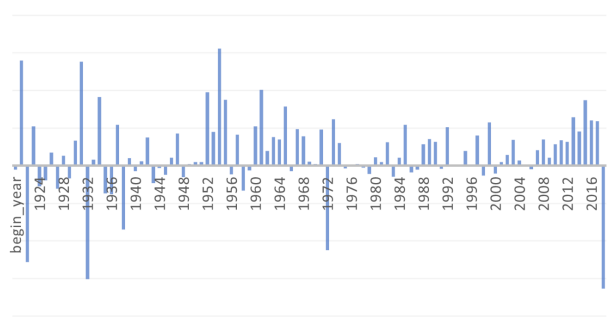

Figure 17: Popularity Change

As for the change of music propensity from 1954 to 1955, the most popular music genre shifted from Vocal to Pop/Rock, which remained to be the most popular genre in the next few decades. Specifically, Pop/Rock suddenly became prevalent between 1955 and 1956, and last for a long time until now, as seen in Figure 18. The specific popularity value is listed in Table 4. In this way, it is easy to relate the increase of popularity of all music to the increase of popularity of Pop/Rock.

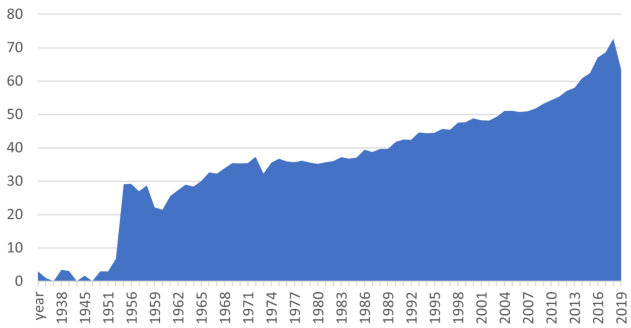

Figure 18: Popularity of Pop/Rock

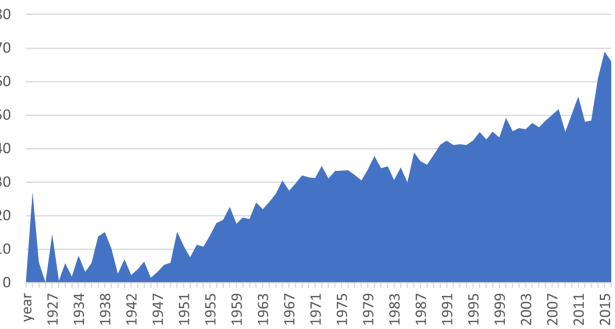

Figure 19: Popularity of Jazz

Table 4: Most Popular Genre from 1953 to 1960

Year	Most Popular Genre	Popularity
1953	Jazz	3.1774
1954	Country	7.0835
1955	Vocal	8.8735
1956	Pop/Rock	15.1040
1957	Pop/Rock	18.6135
1958	Pop/Rock	18.1555
1959	Pop/Rock	19.8050
1960	Latin	18.4845

Moreover, from Figure 14, we can see that artists in the Pop/Rock genre have dimmed the danceability while added instrumentalness to their works at this historical period. Therefore, we can draw a conclusion that the important characteristics that can signify revolution are danceability and instrumentalness.

Next, we will discuss the revolutionary artists. First of all, it is necessary to define what makes a revolutionary. We suggest that these artists should:

- have a certain amount of influence,i.e. this artist possesses a high PageRank;
- make substantial and pioneering contribution.

Here we offer two counter-examples. The first type of artists is the influential Pop/Rock ones, especially Bob Dylan and The Beatles, who are quite in common with a high PageRank and high popularity. However, they are more likely to be the milestone of this genre rather than a significant revolutionary, because during their age, Pop/Rock was already a well-known and highly popular genre. Another example is Cab Calloway. According to our data, many great artists are influenced by him, but the problem is that he did not take Jazz to an unprecedented level. Rather, he just maintained a certain level of popularity.

So we prefer to give the title of "revolutionary" to Chuck Berry— although slightly less influential and popular than the above two examples, he managed to bring Pop/Rock to the public in the 1950s and started the glorious history of this genre.

We also find that among Chuck Berry's influence, Muddy Waters is the second most similar [1] to Chuck Berry, which indicates that Berry's music is heavily influenced by Waters. Interestingly, Waters belonged to Blues in terms of genre, so we make an audacious speculation that Berry absorbed the essence of Blues and improved the Pop/Rock style in a pioneering way, making the development of Pop/Rock enter a new era. In fact, the history is in accordance with our conclusion. Berry was inspired by Blues and developed the elements of Rock n' Roll from charateristics of Blues. [6] Therefore, we regard Chuck Berry as an influential and revolutionary artist.

[1] Here, the similarity refers to the Cosine Similarity. Muddy Waters is the influencer and Berry the follower. Besides, the most similar artist to Berry is Jerry Garcia who is influenced by Berry.

7.3 Examination of External Changes through Music Evolution

The evolution of music, as a major part of the cultural changes, has been particularly evident for decades in the United States. In the recent history of American music, Pop/Rock has always occupied a pivotal position. We have examined the change in popularity of Pop/Rock over the last 90 years and find that its popularity soared from 1951 to 1956, significantly faster than the growth of Jazz, Country, and other genres. Since the nineteenth century, the United States has absorbed a lot of European folk music brought by immigrants and African music brought by blacks, and these diverse styles of music blended together to form a series of novel music. In the evolution of American music in recent decades, the popularity of these emerging music has increased dramatically, with Pop/Rock being the most obvious one, which in turn has led to the evolution of the entire music industry. From Figure 16 (popularity of the whole music industry), it is clear that between 1951 and 1956, music industry was also increasing dramatically, which owes to the most important factor: the popularity of Pop/Rock music. Thus, we can find that Pop/Rock genre has gradually become the biggest influencer in the evolution of American music, bringing about a cultural change in the United States.

Additionally, as we probe deeper into the continuity of the evolution of music, after examining the popularity data set, we come to a conclusion that music in the United States has been constantly evolving in recent times. However, there are two particular periods when music evolved at a particularly fast pace: 1951–1956 and 2008–2012, as shown in Figure 16 and Figure **??**. From previous analysis, much of the growth in popularity from 1951 to 1956 is related to the soaring popularity of Pop/Rock. It seems that the development from Jazz to Pop/Rock is a great success, making Pop/Rock rapidly become the mainstream genre. Moreover, the United States built the industrial system of pop music on a well-defined business model and exported music to all the whole world through a complete system. This cultural infiltration made American music universal. Additionally, 2008 to 2012 was the period when smart phones became prevalent, and this technological change brought about a rapid rise in the popularity of American music.

8 Sensitivity Analysis

Equal proportion reduction and equal compensation

In Directed Network we find that some nodes are very selfish: they constantly absorb the value of others, but do not share it with others.

After running the basic PageRank update rule every time, the PageRank value of each node is multiplied by a scale factor s ($0 < s < 1$, and the empirical value is between 0.8 and 0.9), then add (1-s) / N to the PageRank value of each node.

$$PR(t) = s * PR(t) + \frac{1-s}{N}, \quad 0 < s < 1 \tag{5}$$

In this way, we not only maintain the sum of every node's PageRank value is 1, but also prevent the PR value from being over concentrated to individual nodes.

9 Model Evaluation and Further Discussion

After close examination and evaluation of our models, we put forward some strengths and weaknesses.

9.1 Strengths

- **Comprehensiveness of parameters:** We introduce PageRank(PR), Degree-Centrality(DC), Closeness-centrality(CC), and Betweenness-Centrality(BC) to quantify music influence.
- **Scalability:** Our EMCA framework can be used to analyze the evolution of a bunch of things. In addition to studying the evolution of music, we can also study the evolution of literature, history, etc.
- **Visualization:** We visualize the calculation results in many places, which makes the conclusion more intuitive and increases the interpretability of the model.
- **High Accuracy:** Our model analysis results are in good agreement with the authentic situation, indicating that our model has high accuracy.
- **Innovation:** We design the I index to measure the influence of music, and the D index to measure the appeal of music features.

9.2 Weaknesses

- When the number of music genres becomes large, our Decision Tree Model may need a long training time, and the classification effect is unstable.
- Since we assume that the music influence is instantaneous, we cannot study the music influence with hysteresis effect.

9.3 Further Discussion

Moreover, we develop some further discussion about our models.

In our Cluster Analysis of music, we find that the results show a high degree of consistency with the genre classification. This suggests that it is feasible to classify music based on its music characteristics. Therefore, we propose an idea to classify music based on its music characteristics. Based on this, online music genre can be better classified to have a better music recommendation system and reduce the controversy of genre categorization. Classifying music with Cluster Analysis can improve the rigor of the music industry, but it also faces the problem that some minor music genre will be classified into mainstream music, which might reduce the number of existing genres to some extent.

In our study of the evolution of music, we quantify music characteristics and set specific indicators of music similarity and music influence, which allow us to characterize the evolution of music in terms of the influence of artists on each other and the substitution of genres for each other. However, our study is not limited to the music industry. In fact, our study points the way to a quantitative science of cultural change.

10 Conclusion

In our paper, we establish the EMCA framework to examine evolutionary and revolutionary trends of artists and genres.

First, we use Directed Networks to determine the influence of artists and genres. Secondly, we demonstrate that artists within the same genres are more similar than artists across genres. Also, we verify that there is authentic influence between follower and influencer. Moreover, we analyze the relationship between music characteristics and genres, finding the three most decisive characteristics among multiple music characteristics. In addition, we find that there exists integration of genres, resulting in a high degree of similarity between most genres. Finally, we summarize the basic trends in music from 1921 to 2021, using data to explain the phenomena and draw a conclusion that "music development is a process of constant integration".

11 MEMORANDUM

To: ICM Society

From: Team #2118577

Date: February 9, 2021

Subject: Research Result of Music Influence, Music Evolution, and Cultural Changes

Honorable ICM Society,

In accordance to your requests, we have designed the EMCA framework to explore music evolution and the corresponding cultural changes.

Firstly, we build a Directed Network Model based on the data in *influence_data* data set, which is somewhat similar to a social network model. We apply it to examine the influence of artists and genres, and further identify the most influential artists in the network, including big names like The Beatles.

Then, we analyze the similarity between artists and genres based on their music characteristics respectively. In this case, the similarity of works within the same genre is indeed higher than that between genres, which affirms that there is a certain pattern in the classification of music. Meanwhile, despite the large number of genres, most of them have a high degree of similarity to each other, which provides a basis for further discovery of the patterns of music development. Among artists, 68% of the influences are contagious, meaning that followers does show similarities in music characteristics to their influencers.

In the third section, we explore the relationship between music characteristics and genres, hoping to predict genres by music features. In this process, we identify three music features,namely acousticness, instrumentalness, and danceability, as the main factors influencing musical genres. Jazz, Pop/Rock, Classical, and Vocal are four genres with better result under supervised learning, mainly because these types of music have more obvious characteristics. We construct a huge Decision Tree for analyzing genre classification and figure out the prominent music characteristics.

Finally, we build a Time Series Model. One of the key findings is the seismic shift between 1955 and 1956 when Pop/Rock became the mainstream in the music field, with the lead character being Chuck Berry, the revolutionary artist who successfully integrate Blues into Pop/Rock. After him came some phenomenal figures such as Bob Dylan and The Beatles, and Pop/Rock remains a hitherto high popularity.

Overall, we propose a comprehensive model that not only explains the inherent patterns between genres and music characteristics, influencers and followers, but also reflects historical and cultural changes. From the perspective of a data analyst, we witness the dramatic shift in music features in the 1950s, which took music to an unprecedented level. Afterwords, music is constantly in the process of interaction and integration.

Ultimately, our EMCA framework demonstrates how to predict the evolution of music. It can be applied to make further prediction if data is provided. We hope that our models, results, and suggestions could provide you with valuable information.

Sincerely,

Team #2118577

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Appendices

Due to space limitations, we will provide three main codes from our models.

Appendix A Code for PageRank

```
import networkx as nx
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

center = 13511

def show_graph(graph, layout='spring_layout'):
    if layout == 'circular_layout':
        positions=nx.circular_layout(graph)
    if layout == 'spectral_layout':
        positions=nx.spectral_layout(graph)
    if layout == 'spring_layout':
        positions=nx.spring_layout(graph)
    positions[center] = (0.05,0.1)
    nodesize = [x['pagerank']*40000 for v,x in graph.nodes(data=True)]
    nx.draw_networkx_nodes(graph, positions, node_size=nodesize, node_color='m',
                                                node_shape='o', alpha=0.7)
    nx.draw_networkx_edges(graph, positions, alpha=0.7)
    nx.draw_networkx_labels(graph, positions, font_size=7)
    plt.show()

data = pd.read_csv("influence_data.csv",encoding="utf-8")

influencer = data["influencer_id"]
follower   = data["follower_id"]

x = influencer.values.tolist()
y = follower.values.tolist()

G = nx.DiGraph()

for i in range(len(x)):
    G.add_edge(y[i],x[i])

pr = nx.pagerank(G,alpha=0.85)

with open("pagerank_result.csv","w",encoding="utf-8") as f:
    for node,value in pr.items():
        f.write(str(node)+","+str(value)+"\n")

layout = nx.spring_layout(G)
plt.figure(2)
nx.draw(G, pos=layout, node_size=[x*6000 for x in pr.values()], node_color='m',
                                            with_labels=False)
plt.show()

nx.set_node_attributes(G, name = 'pagerank', values=pr)

small_graph = G.copy()

for n, p_rank in G.nodes(data=True):
    if n==center:
        continue
    if G.has_edge(center,n)==False and G.has_edge(n,center)==False:
        small_graph.remove_node(n)

show_graph(small_graph,'circular_layout')
```

Appendix B Code for Decision Tree

```
from itertools import product
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
import pydotplus
import graphviz

from sklearn import datasets
from sklearn import tree
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier

import os
os.environ["PATH"] += os.pathsep + 'C:/Program Files/Graphviz/bin'

X = pd.read_csv("full_normalize.csv")
y = pd.read_csv("full_genre.csv")

X_train,X_test,y_train,y_test = train_test_split(X,y,test_size=0.2,random_state=
                                        42)

clf = DecisionTreeClassifier(   criterion='gini', splitter='best',
                                max_depth=None, min_samples_split=300,
                                min_samples_leaf=10, min_weight_fraction_leaf=0
                                                                          .,
                                max_features=None, random_state=None,
                                max_leaf_nodes=None, min_impurity_decrease=0.,
                                min_impurity_split=None, class_weight=None,
                                                                     ccp_alpha
                                                                     =0.0
                                 )
clf.fit(X_train, y_train)
print(clf.score(X_test, y_test))

dot_data = tree.export_graphviz(clf, out_file=None,
feature_names=["danceability","energy","valence","tempo","loudness","key",
               "acousticness","instrumentalness","liveness","speechiness",
               "mode","explicit"],
               class_names=["0","1","2","3","4","5","6","7","8","9",
               "10","11","12","13","14","15","16","17","18","19","20"],
                        filled=True, rounded=True,
                        special_characters=True)
graph = pydotplus.graph_from_dot_data(dot_data)
graph.write_pdf("iris.pdf")

y_importances = clf.feature_importances_
x_importances = ["danceability","energy","valence", "tempo","loudness","key",
           "acousticness","instrumentalness","liveness","speechiness",
           "mode","explicit"]
y_pos = np.arange(len(x_importances))
plt.show()
```

Appendix C Code for K-Means

```
import numpy as np
import pandas as pd
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

data_genre = pd.read_csv("full_genre.csv",encoding="utf-8")
genre = data_genre["genre"]

clf = KMeans(n_clusters=3)
clf.fit(x)

centers = clf.cluster_centers_
labels = clf.labels_

mark = ['or', 'ob', 'og', 'ok', '^r',
        '+r', 'sr', 'dr', '<r', 'pr']

data = pd.read_csv("full_normalize.csv",encoding="utf-8")

ac = data["acousticness"]
dc = data["danceability"]

for i in range( len(ac) ):
    if genre[i]==4 or genre[i]==5 or genre[i]==10:
        plt.plot(ac[i], dc[i], mark[int(genre[i])-2] )
plt.show()
```